AI251P Applied Computer Vision

Capstone Project – Phase 1 Report

Camera-Trap Image Classification

Project 5 · Phase 1 (M1–M3, 40 points) · Due 28 October 2026

1 Team and project

Field	Description
Team name	Project 5
Topic	Project 5: camera-trap image classification
Application area	Wildlife monitoring and image triage
Code repository	https://github.com/salehdev3/Capstone_5
W&B project	https://wandb.ai/saleh-alanazi-ministry-of-interior/camera-trap-phase1
Dataset card	https://github.com/salehdev3/Capstone_5/blob/v1/DATASET_CARD.md

Team members

Name	Student ID	Workstream (Table 3)	Phase 1 contribution
Saud Abdullah Alfaiz	225462	Data and classification	Data preparation, grouped split, audit and card
Saleh Mohammed Alanazi	225486	Data and classification	Scratch CNN and evaluation
Yousef Nafi Alanazi	225457	Data and classification	Frozen and full ResNet-50 comparison
Mohannad Mohammed Alqahtani	225461	Integration	Independent review, comparison and report

Problem and acceptance criteria

- **Summary:** The system classifies full camera-trap frames as empty or animal, then predicts animal species on human-box crops. It reports per-class metrics, calibration and errors.
- **Input → output:** A camera-trap photograph → empty/animal decision; an annotated animal crop → species category.
- **Intended users and the decision supported:** Wildlife-data reviewers use it to prioritize animal frames and inspect species predictions.
- **Acceptance criteria (fixed before results):** Animal recall $\geq$ .99; empty removal $\geq$ .20; species macro-F1 $\geq$ .65; species ECE $\leq$ .10.

2 Data foundations (M1) (12 points)

2.1 Datasets

Dataset	Link / ref.	Content and annotations	Use	Licence	Images
CCT20 subset	LILA CCT	Camera-trap frames and human boxes	All splits	CDLA-Permissive	6,022
Team frames	Local collection	Reviewed frames and boxes	Train/test	Team-owned	19

The notebook selected 6,022 CCT20 images under a fixed seed of 42, then included 19 reviewed team images. This project-specific subset does not use the official CCT benchmark split.

2.2 Cleaning

Check	Method / tool	Images removed
Corrupt or unreadable files	Download/decode checks	0
Exact duplicates	ID/hash checks	0
Near-duplicates	pHash cross-partition checks	0
Other (human audit quarantine)	One flagged event, four frames	4
Images remaining	6,041 initially kept; 6,037 after audit	6,037

2.3 Leakage-safe split

- **Grouping key:** CCT20 camera/event; local images by city and session.
- **Why this prevents leakage:** Frames from the same camera or burst share backgrounds, lighting and animals; grouping keeps these correlations within one partition.
- **Stratification and seed:** Group-aware approximate joint-label stratification, seed 42. Camera separation takes priority over exact class proportions.

Split	No. of Groups	No. of Images	Images per class (or % of the smallest class)
Train	13	3,534	Empty: 319; animal: 3,215
Validation selection	3	1,337	Empty: 91; animal: 1,246
Validation calibration	2	527	Empty: 13; animal: 514
Test	4	639	Empty: 53; animal: 586

2.4 Species crop support by split

Species	Train	Selection	Calibration	Test
Badger	4 (0.1%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Bird	116 (4.1%)	3 (0.3%)	45 (9.3%)	9 (1.8%)
Bobcat	286 (10.1%)	25 (2.2%)	78 (16.1%)	57 (11.2%)
Cat	142 (5.0%)	247 (22.0%)	35 (7.2%)	48 (9.4%)
Coyote	288 (10.2%)	29 (2.6%)	37 (7.7%)	169 (33.3%)
Deer	11 (0.4%)	0 (0.0%)	0 (0.0%)	3 (0.6%)
Dog	117 (4.2%)	33 (2.9%)	73 (15.1%)	29 (5.7%)
Fox	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Opossum	622 (22.1%)	489 (43.5%)	99 (20.5%)	87 (17.1%)
Rabbit	317 (11.2%)	164 (14.6%)	61 (12.6%)	4 (0.8%)
Raccoon	709 (25.2%)	14 (1.2%)	28 (5.8%)	28 (5.5%)
Rodent	38 (1.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Skunk	73 (2.6%)	0 (0.0%)	21 (4.3%)	19 (3.7%)
Squirrel	95 (3.4%)	120 (10.7%)	6 (1.2%)	53 (10.4%)
Total crops	2,819 (100%)	1,124 (100%)	483 (100%)	508 (100%)

Each cell gives crop count and percentage of crops in that split. Crop counts differ from frame counts because a frame may contain multiple annotated animals. Species with zero support in a split are shown as zero.

The table reports annotated animal crops, not frame counts. A frame may contain multiple animals. Zero denotes no annotated crop for that species in the corresponding split. The source counts are archived in `evidence/species_crop_support.csv`.

2.5 Label audit

Audited sample size	Errors found	Estimated label-error rate	Upper bound (95% confidence)
10 training events	1	10%	39.4% one-sided

One frame was drawn from each of 10 random training events; the team found one incorrect label and quarantined the affected event. A separate 20-image review signed by Mohannad Mohammed Alqahtani found no additional frame-label errors; five box checks were not applicable.

2.6 Data loader and augmentation

RGB images were prepared with maximum side 1,024 pixels, resized to 224×224, then ImageNet-normalized. Training alone uses random flip and brightness/contrast augmentation from the shared preprocessing module; these variations reflect camera viewpoint and illumination. Exact augmentation parameters are fixed in `preprocessing.py`.

2.7 Dataset card summary

Raccoon and opossum dominate some groups while rare species have few examples; camera-group splits have distribution shift. CCT20 is largely US wildlife, and five held-out Tabuk frames cannot establish Saudi deployment performance. The small audit has a wide error bound. Source

terms and course policy restrict image redistribution. Team images are tracked in the repository and are subject to the team collection and publication policy. The full card is linked in Section 1.

3 Classification (M2–M3) (20 points)

Task: Full frames are classified empty/animal; human-box crops are classified by species. These classifiers provide the baseline for later detection and segmentation.

3.1 Runs compared

Six regimes were compared using validation-selection macro-F1; frozen ResNet-50 was chosen for both tasks. The locked test set was evaluated after selection. W&B links for all six runs are in the linked project.

Run (W&B link)	Backbone	Training	Input res.	Epochs	Val. acc.	Val. macro-F1
Triage (W&B)	CNN	From scratch	224	6	.941	.601
Triage (W&B)	ResNet-50	Pretrained, frozen	224	12	.940	.662
Triage (W&B)	ResNet-50	Pretrained, full fine-tuning	224	12	.927	.640
Species (W&B)	CNN	From scratch	224	12	.505	.156
Species (W&B)	ResNet-50	Pretrained, frozen	224	12	.761	.364
Species (W&B)	ResNet-50	Pretrained, full fine-tuning	224	12	.750	.340

Test-set results (final models only)

Run	Test accuracy (95% CI)	Test macro-F1	ECE
Triage scratch	.917 [.868,.966]	.478	.0463
Triage frozen (selected)	.933 [.897,.967]	.652	.0460
Species scratch	.350 [.284,.386]	.178	.0733
Species frozen (selected)	.742 [.642,.876]	.482	.0568

Which model you keep and why: Frozen ResNet-50 gave the highest validation macro-F1 for both tasks (.662 triage and .364 species). It was therefore retained before test results were examined. At test threshold .3416, triage retained 586/586 animal frames but removed only 6/53 empty frames (11.3%), missing the 20% empty-removal target. Species test macro-F1 was .482, below the .65 target; ECE .0568 met the .10 limit.

All accuracy and per-class tables above use argmax. The separate operating policy uses animal probability ≥ 0.3416247, selected on `val_calibrate` to maximize empty removal subject to empirical animal recall $\geq .99$. Calibration retained 509/514 animals and removed 2/13 empty frames. This empirical constraint is not a population guarantee.

Accuracy intervals are 2.5th–97.5th percentile camera-cluster bootstrap intervals (1,000 resamples, seed 42), resampling whole cameras with replacement. Only four test groups are available; intervals are descriptive and unstable. They are not independent-image confidence intervals.

Species macro-F1 is .481994 across all 14 training classes, and .519070 across the 13 classes with test support. Rodent has zero test examples. Both definitions are retained; excluding the unsupported class does not indicate model improvement.

The held-out Tabuk subset contains five frames and overlaps the combined test. Saved per-camera

argmax accuracy is 2/5 for triage and 4/5 for species. This small, imbalanced subset does not establish deployment performance and is not an additional independent test.

3.2 Learning curves

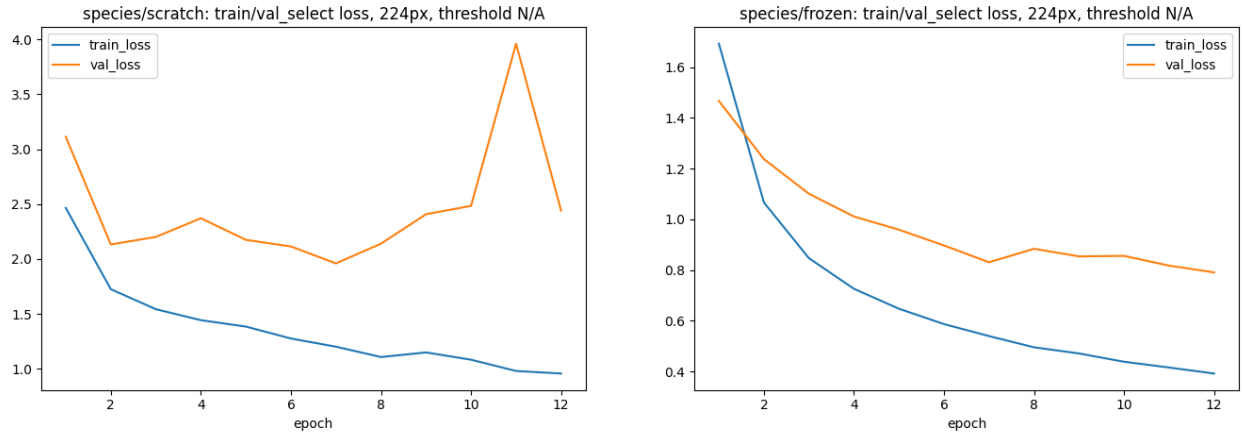


Figure 1: Training and validation-selection loss: (left) scratch CNN, (right) frozen ResNet-50. Split: train and validation-selection; input: 224×224; threshold: argmax.

Species full fine-tuning reached 94.7% training accuracy but 75.0% validation accuracy, consistent with overfitting. Frozen species reached 89.7% and 76.1%, respectively. Rare-class support and camera shift remain important limits.

3.3 Per-class results (selected model, test set)

Class	Precision	Recall	F1	Support
Empty (triage)	.917	.208	.338	53
Animal (triage)	.933	.998	.965	586
Badger (species)	.000	.000	.000	1
Bird (species)	.900	1.000	.947	9
Bobcat (species)	.629	.684	.655	57
Cat (species)	.632	.250	.358	48
Coyote (species)	.956	.763	.849	169
Deer (species)	.000	.000	.000	3
Dog (species)	.676	.862	.758	29
Fox (species)	.000	.000	.000	1
Opossum (species)	.656	.966	.781	87
Rabbit (species)	.250	.250	.250	4
Raccoon (species)	.366	.536	.435	28
Rodent (species)	.000	.000	.000	0
Skunk (species)	.933	.737	.824	19
Squirrel (species)	.860	.925	.891	53
Macro average (species)	.490	.498	.482	508

Top-k accuracy (species, $k = 3$): .9055 on test; 224×224, argmax.

3.4 Triage confusion matrices

Rows are true labels; columns are predictions. Split: test, input: 224 × 224. These tables use all 639 frames.

	Argmax		Threshold .3416247	
True label	Empty	Animal	Empty	Animal
Empty	11	42	6	47
Animal	1	585	0	586

3.5 Confusion matrix and calibration

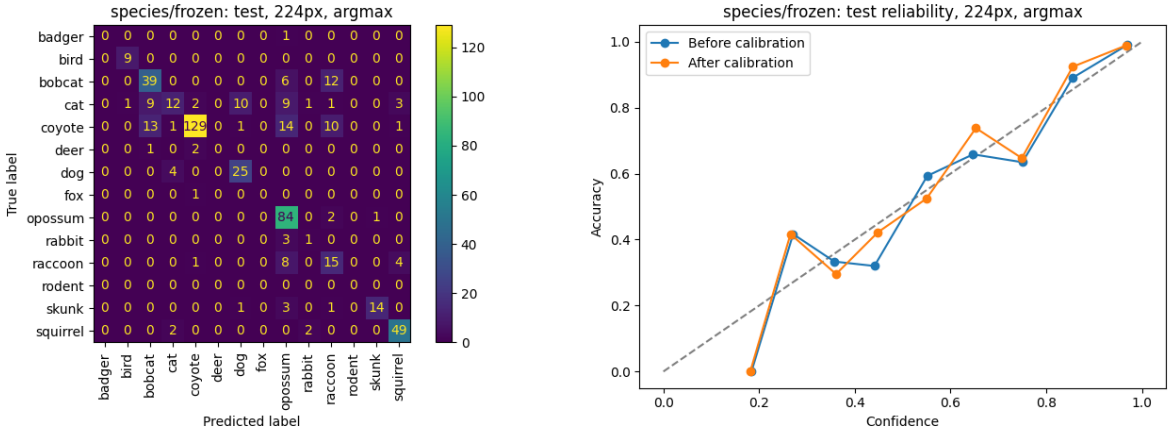


Figure 2: (Left) confusion matrix; (right) reliability diagram. Split: test; input: 224×224; threshold: argmax.

Bobcat was repeatedly confused with raccoon in the gallery; a fox crop was called coyote. Selected species test ECE was .0568 versus raw .0454, so temperature scaling did not improve ECE on this test split. Gallery observations are not a causal proof.

3.6 Error analysis

Error pattern	Full-test count / gallery	Likely cause (data, label, model)
Recorded empty → animal	47 total; 12 shown	Bird-like visual features in several daylight frames indicate potential source-label ambiguity.
Bobcat → raccoon	12 total; 5 shown	Low-detail night crops and camera-domain shift are plausible contributors.

Counts above cover all test decisions: triage uses the calibrated threshold, species uses argmax. Visual explanations are hypotheses, not counts of verified night-time causes.

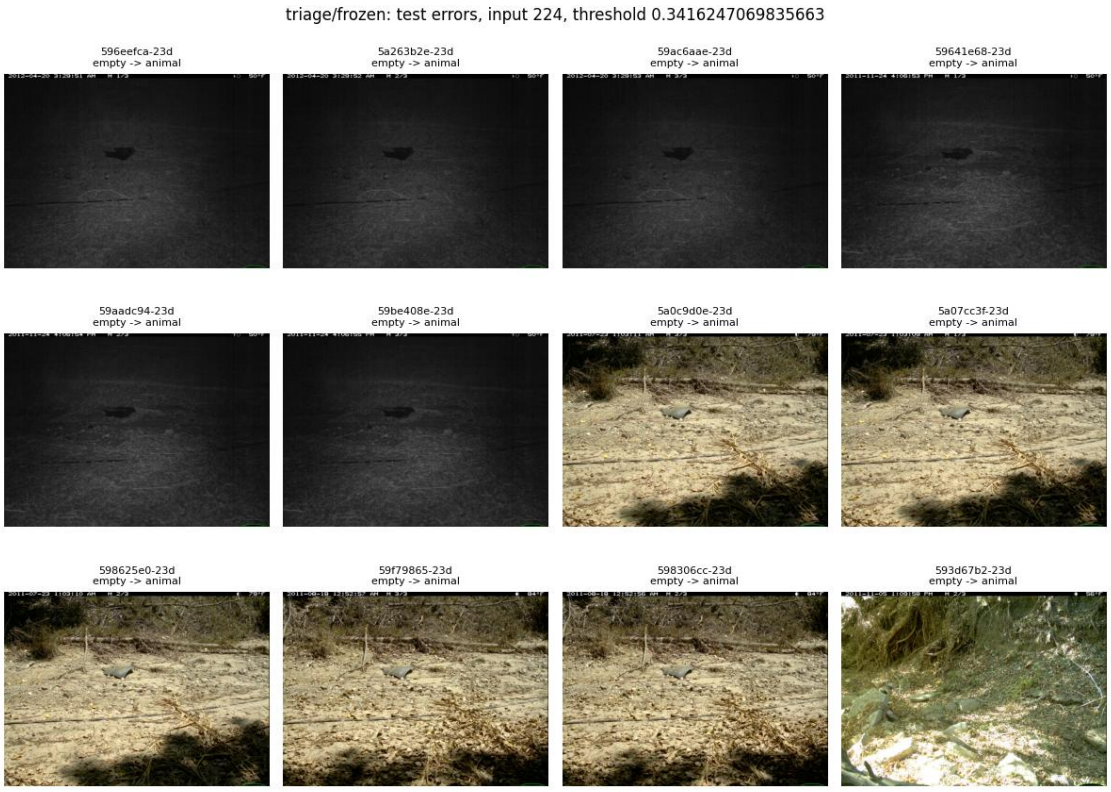


Figure 3: Triage test error gallery: recorded label and prediction are shown; input 224×224, animal threshold .3416. Visual inspection indicates potential source-label ambiguity.

3.7 Data-improvement plan

Finding from the error analysis	Planned action	Expected effect
Possible source-label errors	Full-resolution annotation review and a separate corrected-label sensitivity analysis	Distinguish annotation issues from model errors
Rare species and empty frames	Validation-selection comparison of class-weighted loss and the baseline, with additional independently grouped examples	Measure minority recall and empty removal without tuning on test
Low-detail night crops	Full-test illumination annotation and validation-based comparison of night-focused training data and illumination augmentation	Night-time recall and calibration measured against the baseline

4 Reproducibility (8 points)

Item	Description
Repository and commit hash	https://github.com/salehdev3/Capstone_5, evaluated results release `2cc1731` on `v1`
Configuration file used for the reported runs	`handoff/data_config.json` and `runs/member2,member3/config.json`
Random seeds	42
Requirements file (pinned versions)	Pinned CPU validation environment: `requirements-cpu-validation.txt`. Historical GPU run configurations are recorded in W&B.
Command(s) to reproduce the main result	Run `notebooks/Phase1_Final_Presentation.ipynb` in section order on Colab GPU
Data sources	CCT20 on LILA; team photos tracked in the repository; recorded manifest hash in dataset card
Reproduction evidence and scope	Fresh-process checkpoint inference is recorded in `review/fresh_process_results.json`. Instructor feedback separately identifies a matching full W&B rerun on 28 October, attributed by the team to Mohannad. Checkpoint inference and full training replay are reported as distinct validation procedures

5 Submission checklist

- ☐ Corrupt files and duplicates handled, with counts reported
- ☐ Split is grouped by the source of correlation; the grouping key is justified
- ☐ Label audit reported with a confidence bound
- ☐ Dataset card states licences, limitations and privacy measures
- ☐ Scratch and fine-tuned classifiers compared in W&B
- ☐ Per-class metrics, confusion matrix and calibration (ECE) reported
- ☐ Test set used only for final evaluation, not model selection
- ☐ Data-improvement plan follows from the error analysis
- ☐ Fixed seeds, recorded training configuration, and a pinned CPU validation environment
- ☐ Every figure states split, input resolution and thresholds
- ☐ Team photos are identified as tracked repository content

6 Self-assessment

Criterion	Points	Self-score	Strongest evidence (section)	Known gap
Data foundations and dataset card (M1)	12	11	Cleaning, split, audit, card	Small initial audit
Classification (M2–M3)	20	19	Six runs, metrics and galleries	Two preset targets missed
Reproducibility	8	6	Config, hashes, inference check	CPU environment pinning and checkpoint inference are distinct from full GPU replay
Total	**40**	**36**	Instructor Phase 1 feedback	Phase 1 assessment

7 Obstacles and how they were addressed

A Colab upload stall interrupted evaluation. The evaluation workflow now saves test metrics before a time-bounded W&B upload, so a stalled upload does not block the local results.

Obstacle	Module	Impact on the work	Action taken	Status
Colab upload stalls	M2–M3	Evaluation interruption	Save metrics before bounded W&B upload	Resolved